Monte Carlo Methods and Nuclear Modeling

Introduction to Monte Carlo Modeling
Problem 2

Under the guidance of Professor:
Christoph HARTNACK

ABSTRACT:

The resolution to this problem comprises the development of a MatLab-GNU Octave script implementing a Monte-Carlo simulation algorithm to simulate the tri-dimensional irradiation of a cylinder containing a homogeneously-dissolved β^--emitter, with mean free path λ=2[mm], assumed to occupy 100% of the cylinder volume.

Multiple iterations of the script for different cylinder heights confirmed the occurrence of the "Infinite Thickness Principle"[5], critically to assure the correct and safe use of radiation detection devices. Refer to "Conclusions" section for concrete warnings and advise.

The script leverages the "Exponential Attenuation Principle" [6][7] from dosimetry cohorts to approximate radiation distribution in real life. It also prevents spherical "Pole Clumping" of data points by implementing course work proprietary mathematical modeling. Finally, and where applicable, results are expressed in terms of the "Cuadrature Principle" [4], also a subject of study from statistics cohorts, to show the accuracy of the script reported values.

Date	Peer 1	Peer 2	Peer 3
02 Feb 2026	Javier Alonzo ROMO LEON	N/A	N/A

The present document formatting has been aligned to meet standard practices as per ISO9001 in support of future standardization processes.

Table of Contents.

1. Introduction.

The present document compiles class exercises as taught by lead “Monte Carlo Methods and Nuclear Modeling” teacher, Professor Christoph HARTNACK, with the intent of:

- Documenting resolution methodology,
- Annotate professor-student comments,
- Document lessons learned,
- Document relevant bibliography,
- Provide a quick entry point to solve related scenarios in future professional endeavors.

1.1. Problem summary and location of solutions.

The problem prompts to find a minimum viable algorithm, preferably MatLab - Octave based, that would allow to implement Monte-Carlo analyses to estimate the efficiency of the detection of a dissolved β-emitter as a function of its container depth. Definitive data and plots are provided in 3.1.

1.2. MatLab – GNU Octave script source code location and communications.

The MatLab - GNU Octave script developed to resolve this Problem 2, is publicly available at: https://raw.githubusercontent.com/J4v1987/MatLab-Octave_Nuclear_Modelling/refs/heads/main/Problem2.m

This MatLab – GNU Octave script has been developed in Ubuntu Linux and requires terminal command.

Should the formatting of figures in this report exhibit suboptimal quality, do consider consulting flat 2D *.png, as well as interactive 3D *.ofig, from the same script project repository:
https://github.com/J4v1987/MatLab-Octave_Nuclear_Modelling/tree/f0c0e29f804f6fefce95caec1312562068b02f74/Problem2_plots

Professionals in the nuclear engineering realm and interested audience are invited to participate in the development of this script. Do consider contacting the repository maintainer via dedicated communications found in:
https://sites.google.com/view/b-eng-jarl/home

1.3. Syntax and key definitions.

camelCaseSyntax denotes script variables.

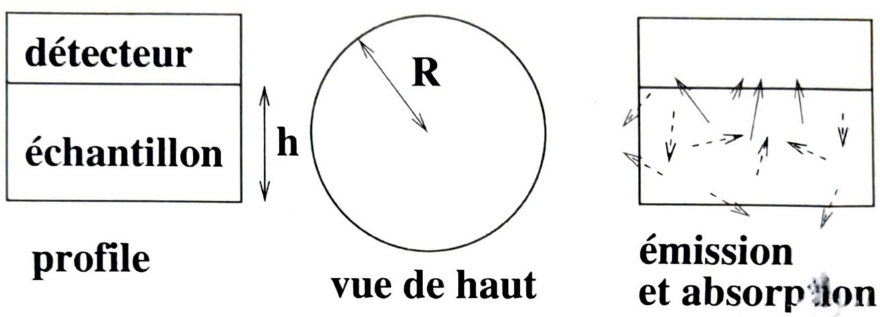
2. Problem 2 Enunciation

Introduction to Nuclear Modelling, Master SNEAM

Student laboratories Christoph Hartnack

Problem 2 : Efficiency of the detection of a dissolved beta-emitter **(evaluated)**

A small cylindical container (radius ***R***=9mm) is containing an dissolved beta-emitter. A detector plate is placed ontop the container to measure the emitted electrons. It is assumed to be perfect and has the same radius as the container. The electrons are emitted isotopically in the whole container. However there mean free path is assumed to be **λ**=2mm.



1. Calculate the efficiency of the detection assuming the container to have a height of **h**=5mm.
2. Trace the detection efficiency as a function of the height of the container
3. Assuming the concentration of the dissolved emitter to be homogeneous and constant independently of the height of the container, the absolute emission yield is thus directly proportional to the volume and thus to the height of the container.
   The detection yield should thus correspond (up to a factor related to the activity) to the product of the efficiency with the volume. Trace this product as function of the height of the container. What do you observe ?

Simulate the system by distributing the points inside the cylinder with a random isotropic emission, estimate whether it would be capable to reach the detector taking into account the mean free path. The latter can be simulated in two ways, either by applying an exponential weight using the distance of the emission point to the detector or by assigning to each particle a randomly chosen mean free path.

Bonus question :
how would the efficiency and the detection yield change, if the emitted particle was a positron, that annihilates with a mean free path of **λ**=2mm ? The annihilation would create a gamma-pair that is emitted isotropically but back-to-back.

Figure 1: Problem enunciation.

2.1. Script Documentation & Problem2 Interpretation

[Statement 1] A MatLab - GNU Octave script is to be produced. E.g. https://raw.githubusercontent.com/J4v1987/MatLab-Octave_Nuclear_Modelling/refs/heads/main/Problem2.m

[Statement 2] There is a cylinder, completely filled with a β^--emitter medium, uniformly dissolved, and exhibiting isotropic emission of electrons, e^-. Its *meanFreePath*, λ, is the "...average distance...(these) travel in the absorber (medium) before interacting..."*[5],* to be expressed in millimeters, [mm] [1] and user-entered when prompted.

[Statement 3] The script assumes the cylinder is in euclidean standard space, E^n [8], featuring a radius (*cylinderRadius*), ρ [mm] [1], "...with axis given by the line segment (of length, H, denoting the cylinder height) with endpoints (x_1,y_1,z_1) and (x_2,y_2,z_2)..."[11], such that z-coordinates project along Z^+ (z-positive axis), explicitly:

1. $x_1=0$, $y_1=0$, $z_1=0$,
2. $x_2=0$, $y_2=0$, $z_2=$*cylinderHeight*.

Both *cylinderRadius* and *H* are to be entered by the user when prompted.

The aforementioned results in a cylinder volume, V_a (*analyticalVolume*) given by the expression $V_a=\pi\cdot\rho^2\cdot H$.[11]

[Statement 4] The script additionally takes-in two more variables from user input. Namely:

1. *N*, expressed in number of point particles, referring to the number of electrons scattered by the β^--emitter solution in a very small fraction of a time.
2. *stat_N*, expressed in number of Monte Carlo iterations, I_{mc}, each of which the script uses to test different spatial distributions of N to estimate statistical metrics, to be described in further detail along this reference guide.

[Statement 5] The *N* variable is implemented in loop "for i=1:N;". This loop iterates *N* times to form matrix *volumePoints* of random $p^1(x,y,z)$ points evenly filling the cylinder volume.

The randomness of each coordinate in every p^1 is assured by linking them to a GNU Octave built-in rand() instance [10] which produces a statistically random number between 0 and 1.

Then, this number is scaled to cylindrical volume proportions depending on the p^1 coordinate being produced. The methodology is proven to work as intended as shown in Figure 2.

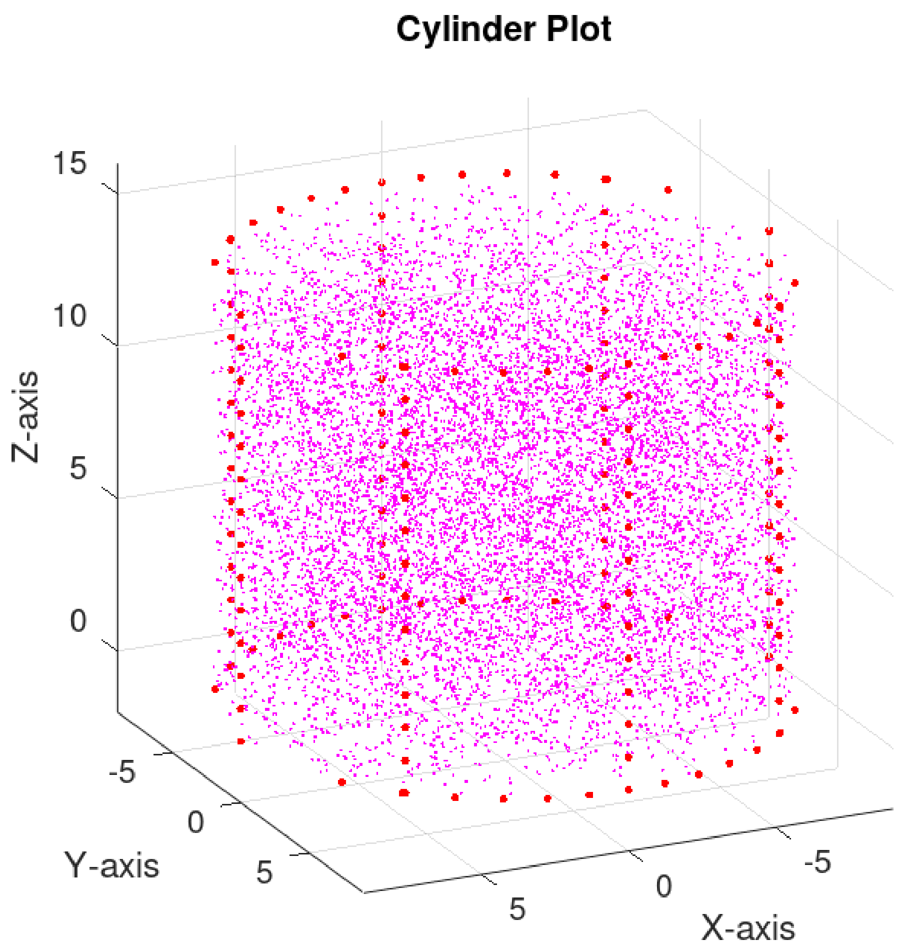


Figure 2: Problem2-MCpoints-h14.png: random points evenly distributed within a cylindrical volume.

[Statement 6] A subsequent loop, "*for i = 1:length(volumePoints)*", assumes each *volumePoints* point recoils an electron with full spherical probability distribution, resulting in a matrix of as many random $p^2(x,y,z)$ points, referred to as matrix *radiationPoints.*

Each *radiationPoints* point represents a p^1 mathematically weighted to match the following principles:

The “**Exponential Attenuation Principle**”[6][7], where the ratio of intensities, $I(x)/I_0$, decay logarithmically as irradiation with mean free path[6], $\lambda=1/\mu_\ell$ [mm], traverse a medium of thickness, L [mm], such that $I(x)=I_0 \cdot e^{-L\cdot(1/\lambda)}$. This leads to:

The “**Infinite Thickness Principle**”[5] where “...If the depth of the sample along the collimation axis is much larger than the mean free path (λ)...in the sample material, all samples of the same physical composition would present the same visible volume to the detector…”[5] mathematically identified as “...7 mean free paths, the distance for which the error in assuming infinite-sample size is less than 0.1%…”.[5]

The sphere “**Pole Clumping**” of points. Such phenomenon is prevented by uniformly weighting each *radiationPoints* point p^2 through the polar angle, ϑ, that is from pole-to-pole, as demonstrated in Appendix A.

[Statement 7] Based on problem enunciation as shown in Figure 1, the cylinder features a particle detector coincident with the cylinder end-point (x_2,y_2,z_2) and perpendicular to its central axis as described in the context of [Statement 3].

Then, the loop “for i=1:length(*radiationPoints*)” generates a new matrix *detectedPoints* to count the number of *radiationPoints, N_e,* that are found above cylinder end-point p^2 coordinate z_2 and within the cylinder circular projected area along z-axis, after the execution of the loop “*for i = 1:length(volumePoints)*”.

[Statement 8] Actions described in [Statement 5], [Statement 6], and [Statement 7], are then repeated a number of Monte Carlo iterations, I_{mc} (script user input: *stat_N*) within the loop “for i=1:*stat_N*”. Each I_{mc} event counts the number of instances temporarily stored in *detectedPoints*, and then lists this number into another matrix called *listOfDetectedPoints* whose sum of each element is stored in *sumDetectedPoints*.

[Statement 9] The following script matrices and variables are then implemented during runtime to perform the calculations prompted in problem enunciation as shown in Figure 1: radiationPoints*, detectedPoints, listOfDetectedPoints, analyticalVolume, N, stat_N,* and *meanFreePath*. These calculations derive in the reported values in script section “*%[TERMINAL REPORT SECTION]*” and “*%[EXTERNAL RAW DATA]*”, where the former produces a GNU Octave terminal output, whereas the latter produces file “Problem2.csv” listing all user prompted results.

The number of detected electrons is reported as $\mu(N_e)$, and it refers to the statistical mean[2] after a number of Monte Carlo iterations, I_{mc}, of the total number of points entered by the user, *N*, for every user-prompted Item. $\mu(N_e)$ is calculated in the script in variable *meanDetectedPoints*=sumDetectedPoints/length(*listOfDetectedPoints*).[9]

As such, $\mu(N_e)$ will feature a tolerance, $\pm\sigma_e$ [e^-], given by the statistical standard deviation (<u>population standard deviation</u>)[3]. σ_e is calculated within the script and retained in variable *standardDeviationOfDetectedPoints* within the context of script section “%Standard deviation of detected points, $\sigma(N_e)$”

In conformance with problem prompts detailed in Figure 1, the script calculates the *detectionEfficiency*, η_e, for every user-prompted Monte Carlo analysis Item, as shown in Table 1, which is to feature a population standard deviation error, σ_p, whose calculation is based on the **Cuadrature Principle** [4] with variables detailed in Appendix B.

Finally, the script calculates the yield, *Y*, as prompted in problem enunciation Figure 1, as directly proportional to η_e and V_a. In the script, this is executed as *detectionYield*=detectionEfficiency*analyticalVolume.

[Statement 10] The script additionally creates, just before end of runtime, the following outputs in the same location where the script is stored and executed:

A *.csv file, Problem2.csv, where each line represent an I_{mc} Item, each separating the following script variables and calculations with commas:
N[particles], I_{mc}[iterations], λ[mm], ρ[mm], H[mm], V_a[mm³], $\mu(N_e)$[particles], $\pm\sigma_e$[mm³], η_e, $\pm\sigma_p$, Y [mm³].

A tri-dimensional plot showing the cylinder and fitting of particles to V_a before scattering, as per last I_{mc} Item, in fixed “*.png” format, Problem2-MCpoints.png (e.g. Figure 2), and an interactive “*.ofig” format, Problem2-MCpoints.ofig.

A tri-dimensional plot showing the cylinder and particle detection scattering from last I_{mc} Item, in fixed “*.png” format, Problem2-3Dplot.png (e.g. Figure 5), and an interactive “*.ofig” format, Problem2-3Dplot.ofig.

2.2. MatLab - GNU Octave script usage in terminal.

The GNU Octave terminal snippet below exhibits a typical usage case of the script.

```
>> clear all
>> close all
>> Problem2
Enter cylinder radius in [mm], ρ: 9
Enter cylinder height in [mm], H: 14
Enter the medium mean free path [mm], λ: 2
Number of Monte Carlo trials per iteration (e.g. 10000), N: 10000
Number of Monte Carlo iterations (e.g. 100), Iₘ꜀: 100
Do you want to generate a 3D plot? (may take longer to process)(yes or no) yes


PROCESSING...


[00. KEY INPUT PARAMETERS & VARIABLES]
Number of scattered electrons from β⁻-decay (user input), N: 10000.00 [e⁻]
Cylinder radius, ρ: 9 [mm]
Cylinder height (entirely filled), H: 14 [mm]
Analytical cylinder volume, Vₐ: 3562.56607 [mm] (∝π·ρ²·H, see: https://mathworld.wolfram.com/Cylinder.html)

[01. DETECTION EFFICIENCY]
Number of Monte Carlo iterations, Iₘ꜀: 100.00 [e⁻]
Mean of the number of detected electrons from β⁻-decay, μ(Nₑ): 290.99000±17.21424[e⁻]

∵ "Detection efficiency", ηₑ, is understood as the mean of the number of detected electrons from β⁻-decay, μ(Nₑ), compared to the total number of emitted electrons, N.
→ ηₑ = μ(Nₑ)/N
→ ηₑ = (290.99000[e⁻])/(10000.00000[e⁻])
⟹ ηₑ = 0.02910±0.00172

∵ "Detection yield", Y, is understood as the arithmetic product of detection efficiency, ηₑ, and analytical cylinder volume, Vₐ.
→ Y = ηₑ·Vₐ
→ Y = (0.02910[e⁻])·(3562.56607[e⁻])
⟹ Y = 103.66711
```

3. Solutions to Problem 2

3.1. Detection Results: η_e (efficiency) and Y(yield)

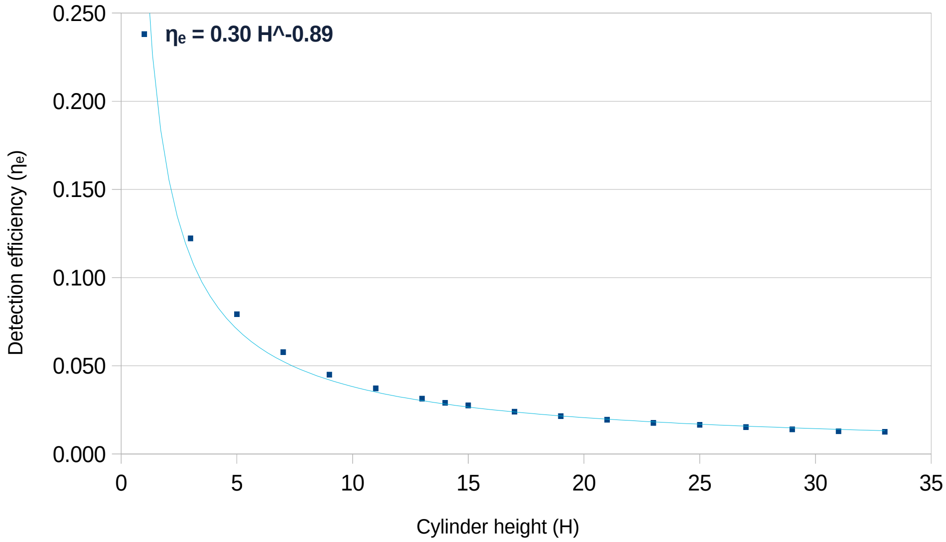
The script was executed 18 times:

- *cylinderHeight was increased* in 2[mm] on every execution.
- N, λ, and ρ were kept the same through all executions.
- Each execution featured 100 Monte-Carlo iterations, I_{mc}.
- Each I_{mc} handled 10000 e^- points, N.

These parameters resulted in Table 1, from which characteristic curves η_e-H (Figure 3) and - Y-H (Figure 4) have been generated. Additionally, a tri-dimensional plot of the cylinder e^- emissions is provided in Figure 5. For more plots, static (*.png) and interactive (*.ofig) are provided in the script project repository (refer to 1.2).

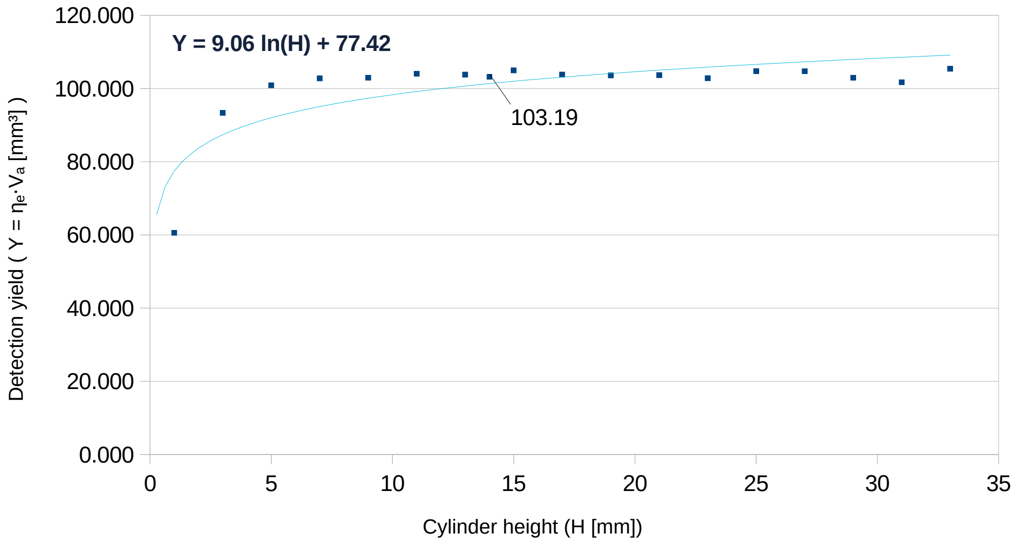
Item	N [points]	I_{mc} [iterations]	λ [mm]	ρ [mm]	H [mm]	V_a [mm³]	$\mu(N_e)$ [nº e^-]	$\pm\sigma_e$ [e^-]	η_e	$\pm\sigma_p$	$Y = \eta_e \cdot V_a$ [mm³]	$\Delta_{mix} = Y/(\pi\cdot\rho^2)$ [mm]	Infinite Thickness [λ]*
1	10000	100	2	9	1	254.47	2,380.38	44.64	0.238	0.00446	60.57	0.24	0.50
2	10000	100	2	9	3	763.41	1,222.50	30.82	0.122	0.00308	93.33	0.37	1.50
3	10000	100	2	9	5	1,272.35	792.82	23.11	0.079	0.00231	100.87	0.40	2.50
4	10000	100	2	9	7	1,781.28	576.96	22.21	0.058	0.00222	102.77	0.40	3.50
5	10000	100	2	9	9	2,290.22	449.52	22.35	0.045	0.00224	102.95	0.40	4.50
6	10000	100	2	9	11	2,799.16	371.63	19.29	0.037	0.00193	104.03	0.41	5.50
7	10000	100	2	9	13	3,308.10	313.82	16.36	0.031	0.00164	103.81	0.41	6.50
8	10000	100	2	9	14	3,562.57	289.66	16.59	0.029	0.00166	103.19	0.41	**7.00**
9	10000	100	2	9	15	3,817.04	274.95	15.01	0.027	0.00150	104.95	0.41	7.50
10	10000	100	2	9	17	4,325.97	239.90	12.56	0.024	0.00126	103.78	0.41	8.50
11	10000	100	2	9	19	4,834.91	214.18	13.25	0.021	0.00133	103.55	0.41	9.50
12	10000	100	2	9	21	5,343.85	193.95	13.51	0.019	0.00135	103.64	0.41	10.50
13	10000	100	2	9	23	5,852.79	175.64	12.77	0.018	0.00128	102.80	0.40	11.50
14	10000	100	2	9	25	6,361.73	164.66	13.95	0.016	0.00140	104.75	0.41	12.50
15	10000	100	2	9	27	6,870.66	152.43	13.02	0.015	0.00130	104.73	0.41	13.50
16	10000	100	2	9	29	7,379.60	139.48	12.17	0.014	0.00122	102.93	0.40	14.50
17	10000	100	2	9	31	7,888.54	128.91	11.90	0.013	0.00119	101.69	0.40	15.50
18	10000	100	2	9	33	8,397.48	125.50	11.91	0.013	0.00119	105.39	0.41	16.50
19	* "…Defined as 7 mean free paths, the distance for which the error in assuming infinite-sample size is less than 0.1%…". Reilly, D. (p. 198-199) https://doi.org/10.2172/5428834 [5]												

Table 1: Problem 2 results. Infinite Thickness Principle check as per [5].



Note:
Each point features a population of 100 iterations.
Each iteration assumed 10,000 electrons had scattered at the same instant.

Figure 3: Efficiency (η_e) as a function of cylinder height (H), from Table 1.



Note:
Each point features a population of 100 iterations.
Each iteration assumed 10,000 electrons had scattered at the same instant.

Figure 4: Yield (Y) as a function of cylinder height (H), from Table 1.

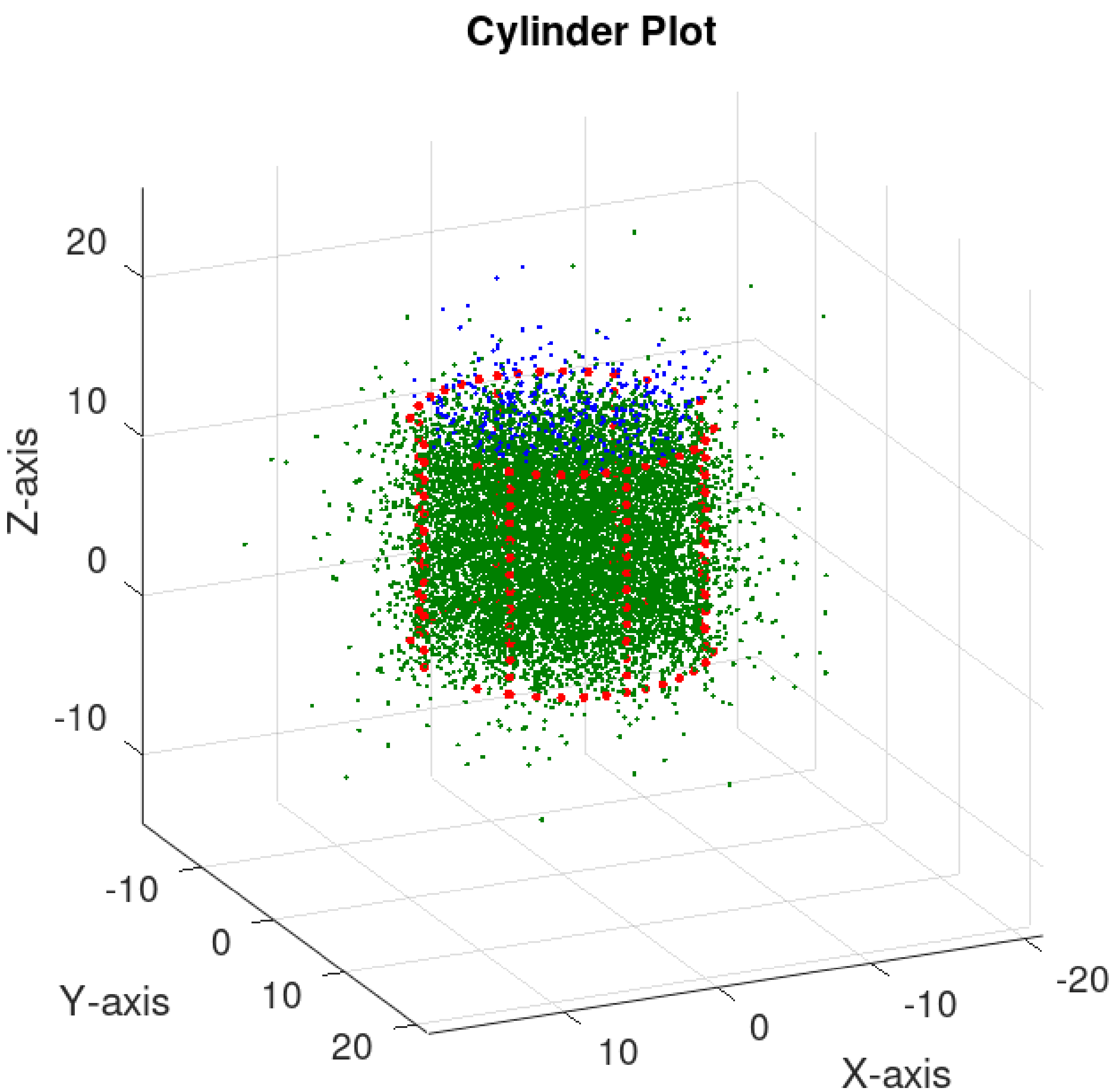


Figure 5: Problem2-h14.png: cylinder e⁻ radiation in green and blue "particles". Blue points refer to the detected particles. The visual count of blue points is noticeably small compared to the green emitted points beyond the cylinder in contrast to Figure 6. This for a cylinder of height 14 [mm] (at Infinite Thickness Principle threshold).

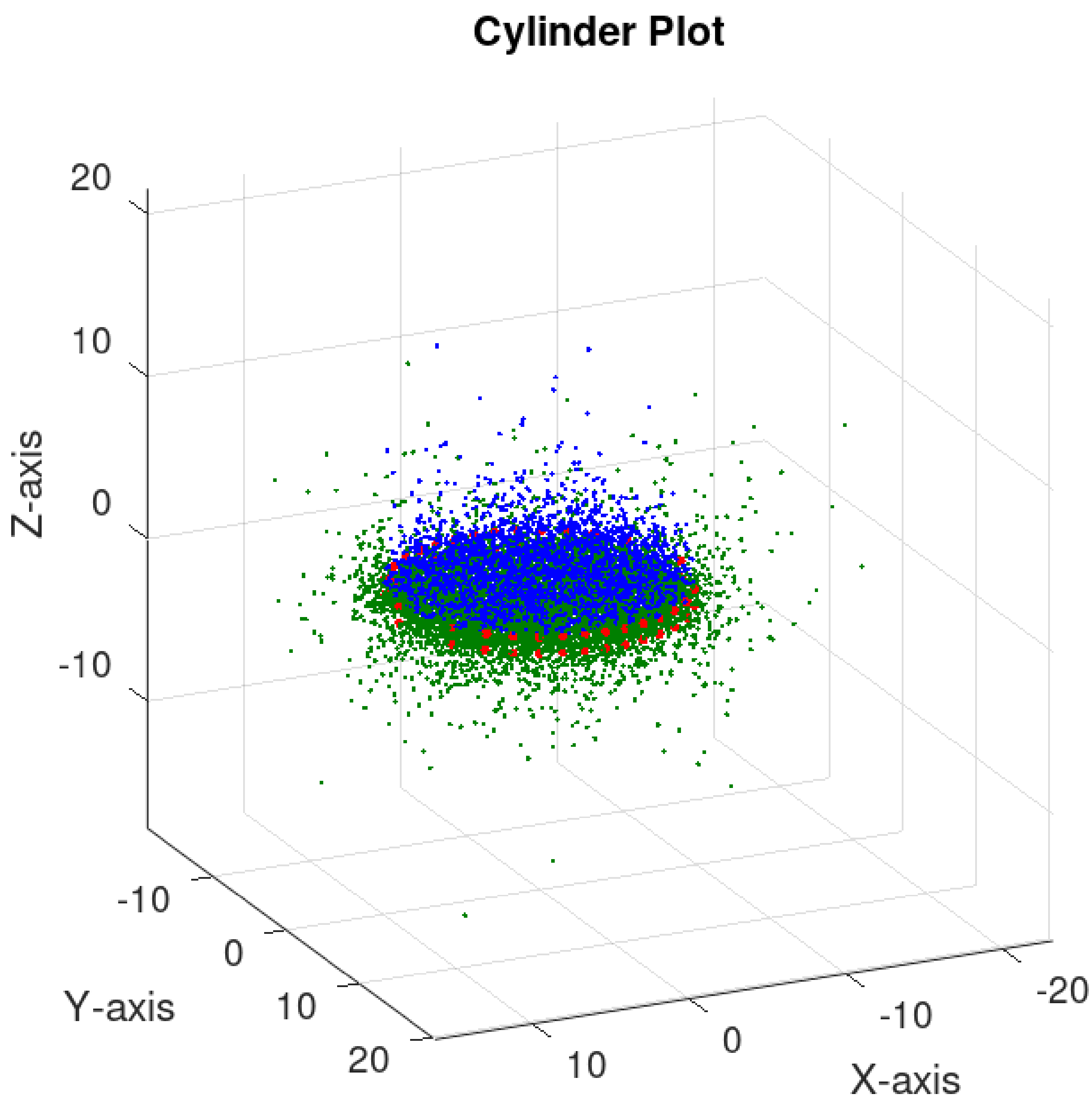


Figure 6: Problem2-h1.png: cylinder e⁻ radiation in green and blue "particles". Blue points refer to the detected particles. The visual count of blue points is noticeably large compared to the green emitted dots beyond the cylinder. This for a cylinder of height 1 [mm].

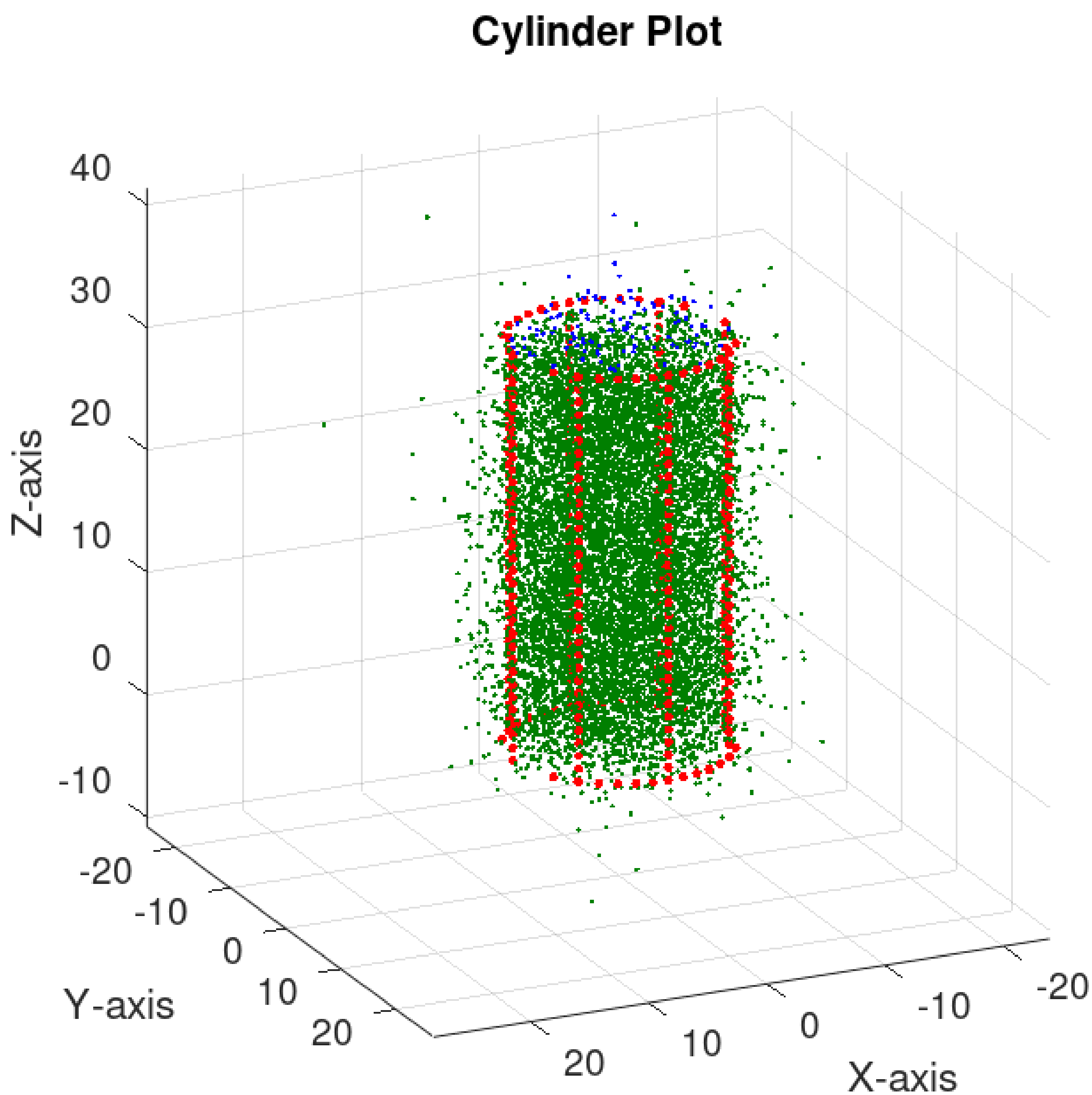


Figure 7: Problem2-h33.png: cylinder e⁻ radiation in green and blue "particles". Blue points refer to the detected particles. The visual count of blue points is noticeably small compared to the green emitted dots beyond the cylinder. This for a cylinder of height 33 [mm].

4. Conclusions

[Conclusion 1] The detection efficiency (η_e) may be related to the cylinder height (H) via equation:

$$\eta_e \approx 0.3 \cdot H^{-0.89} \text{ (note: at best 0.238, or 23.8\%)}$$

As portrayed in Figure 3. Notably, as the height increases, the efficiency decreases.

[Conclusion 2] The detection yield (Y) may be related to the cylinder height (H) via equation:

$$Y \approx 9.06 \cdot \ln(H) + 77.42$$

As portrayed in Figure 4. Notably, as the cylinder height increases, the yield also increases…

[Conclusion 3] ...however, the detection yield, Y, appears to reach a plateau at H ≈ 14 [mm], corresponding to a Y ≈ 103.19 [mm], noted in Figure 4. When looking up these values in Table 1, it is found that this intersection corresponds to ≈ **7.00 [λ]**. This match the **“Infinite Thickness Principle”**[5] as defined by Los Alamos National Laboratory [5].

[Conclusion 4] As portrayed in Figure 4, the **Infinite Thickness Principle** parameter defined as **7.00 [λ]** means that no matter how much more the cylinder height H increases (as described in [Statement 2] and [Statement 3]) to accommodate ever more radiation emitting substance, this will be counter-productive to metrology accuracy as the detection efficiency will start to decrease.

[Conclusion 5] This is because, as the cylinder height H increases, the radiated particles from the bottom of the cylinder would need to travel a longer distance to reach the detector on top of the cylinder, some of which would be absorbed in the irradiating medium before ever reaching the detector.

[Conclusion 6] Therefore, **if H $\gtrapprox$ 7.00 [λ]** as seen in the evolution of Monte Carlo plots, e. g. Figure 6 - Figure 5 - Figure 7, **then the detector efficiency will drop very near η_e=0**. Such condition is **to be averted in real life**, **for it would cause a deceptive ‘safe’ detection count** when in reality **the detector would be omitting to count radiation particles** as noted in [Conclusion 5].

[Conclusion 7] The experiment documented hereby could be further improved in quantitative accuracy by considering more mean free paths, λ, matching the cylinder material and the air around the cylinder, in addition to λ_{medium}. Also a larger I_{mc} would make $\sigma_e, \sigma_p \rightarrow 0$.

5. References

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[2] Illowsky, B. and Dean, S. (2015a). 2.6: Measures of the Center of the Data. [online] Statistics LibreTexts. Available at: https://stats.libretexts.org/Bookshelves/Introductory_Statistics/Introductory_Statistics_1e_(OpenStax)/02%3A_Descriptive_Statistics/2.06%3A_Measures_of_the_Center_of_the_Data See: context & formula 2.6.1.

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[12] Weisstein, E.W. (n.d.). Pythagorean Theorem. [online] mathworld.wolfram.com. Available at: https://mathworld.wolfram.com/PythagoreanTheorem.html [Accessed 30 Jan. 2026].

Appendix A. Derivation of Isotropic Emission Parameters

Goal: Find the probability distribution for the polar angle ϑ (from pole-to-pole) to ensure uniform area coverage on a sphere (Isotropy), while considering a right-handed spherical coordinate system as defined in ISO 80000-2:2009(E). [8] and seen in Figure 8.

1. Differential Area Element (dA):
   ∵dA = (R · dϑ) · [(R · sinϑ) · dφ] [8]
   ∴dA = R^2 · sinϑ · dϑ · dφ
   The probability density function (PDF) is thus proportional to sinϑ.

2. Normalization of PDF (Weighting factor W):
   Let P(ϑ) = ∫ W * sinϑ dϑ over the interval [0, π].
   ∵ 1 = W * ∫[0,π] sinϑ dϑ
   → 1 = W * [-cosϑ][0,π]
   → 1 = W * [(-(-1)) - (-1)]
   → 1 = W * (2)
   ∴ W = 1/2

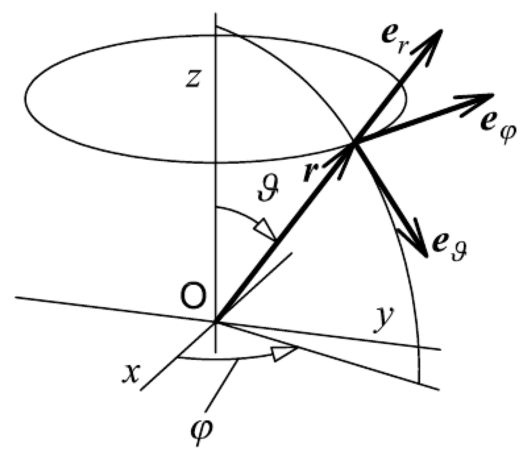

Figure 8: Right-handed spherical coordinate system. [8]

3. Cumulative Distribution Function (CDF):
   Let u be a uniform random number u ∈ [0, 1].
   ∵ u = ∫[0,ϑ] (1/2) * sin(t) dt
   → u = (1/2) * [-cos(t)][0,ϑ]
   → u = (1/2) * [1 - cosϑ]
   → 2u = 1 - cosϑ
   ∴ cosϑ = 1 - 2u
   (Note: Since u is uniform [0,1], 1-2u is statistically identical to 2u-1)

4. Pythagorean Identity for sinϑ:
   ∵$\sin^2$ϑ + $\cos^2$ϑ = 1 (Pythagorean theorem[12])
   ∴ sinϑ = sqrt(1 – $\cos^2$ϑ)

Appendix B. Statistical Error Propagation

From the **Cuadrature Principle**: "...Relative errors add in quadrature." [4]
Let:

$\mu(N_e)$: mean of detected e^- points (or, script variable: *meanDetectedPoints*).
μ_j: mean of total emitted e^- points (or, user input constant, N).
σ_e: standard deviation of detected e^- points (script variable: *standardDeviationOfDetectedPoints*).
σ_j: standard deviation of total emitted e^- points, theoretically 0 because of user input N.
σ_p: standard deviation of η_e referring to the ratio of detected e^- points, and total e^- points given by length (*listOfDetectedPoints*)[9]. Calculation temporarily retained in script variable: *propagatedStandardDeviation*).
η_e: ratio of detected e^- points and total e^- points (or, script variable: detectionEfficiency). Arithmetically: $\eta_e=\mu(N_e)/\mu_j$.

$\because\ \sigma_p/\eta_e=\mathrm{sqrt}(((\sigma_e/\mu(N_e))\text{^}2)+((\sigma_j/\mu_j)\text{^}2))$ (the **Cuadrature Principle**)
$\wedge\ \sigma_j=0$ (because of user input constant, N)
$\rightarrow\ \sigma_p/\eta_e=\mathrm{sqrt}(((\sigma_e/\mu(N_e))\text{^}2)+((0/\mu_j)\text{^}2))$
$\rightarrow\ \sigma_p=\sigma_e\cdot(\eta_e/\mu(N_e))$
$\wedge\ \eta_e=\mu(N_e)/\mu_j$
$\Rightarrow\ \sigma_p=\sigma_e\cdot(1/\mu_j)$